Worksheet 2-7 Derivatives of Important Functions #1
AP Calculus Name_____

Directions: Find the derivative of the following functions.

1. $f(x) = 3\cos(x)$

2. $h(x) = \ln(x) - 4e^x$

3. $f(x) = \sin(x) - \cos(x) + 3x$

4. $g(x) = \dfrac{e^x}{5} - 2\ln(x)$

5. $y = \dfrac{1}{x} - \dfrac{1}{\sqrt{x}} + \dfrac{\cos(x)}{7}$

6. $f(x) = x\sin(2) + 2x^{-5}$

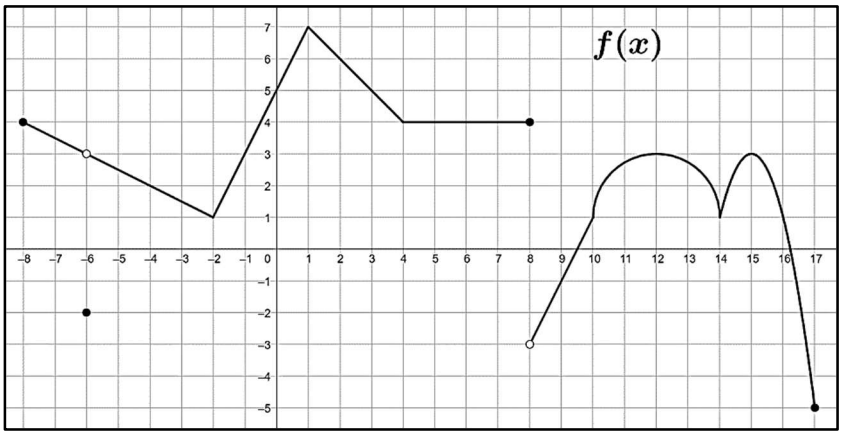
7. $y = \dfrac{3}{x^4} - \sqrt{x} + 4\sqrt{x}$

8. $y = \ln(x) - 2e^x - 4\sin(x)$

Directions: Find the equation of the tangent line at the given x-value

9. $f(x) = 3\cos x + x$ at $x = \dfrac{\pi}{2}$

10. $f(x) = 4e^x - 3\sin x + x^2$ at $x = 0$



Directions: For 11 – 14, the graph of $f(x)$ is above and consists of several segments and a semi-circle on the interval $[-8, 14]$. On the interval $[14, 17]$, f is defined as the equation $f(x) = -2x^2 + 60x - 447$.

11. $h(x) = -2\cos(x) - f(x)$
    Find $h'(3)$

12. $h(x) = e^x + 2f(x)$
    Find $h'(0)$

13. $h(x) = \sin(x) + 3\ln(x) + 3x - f(x)$
    Find $h'(4)$

14. $h(x) = f(x) - 3 + \dfrac{2}{x}$
    Find $h'(16)$